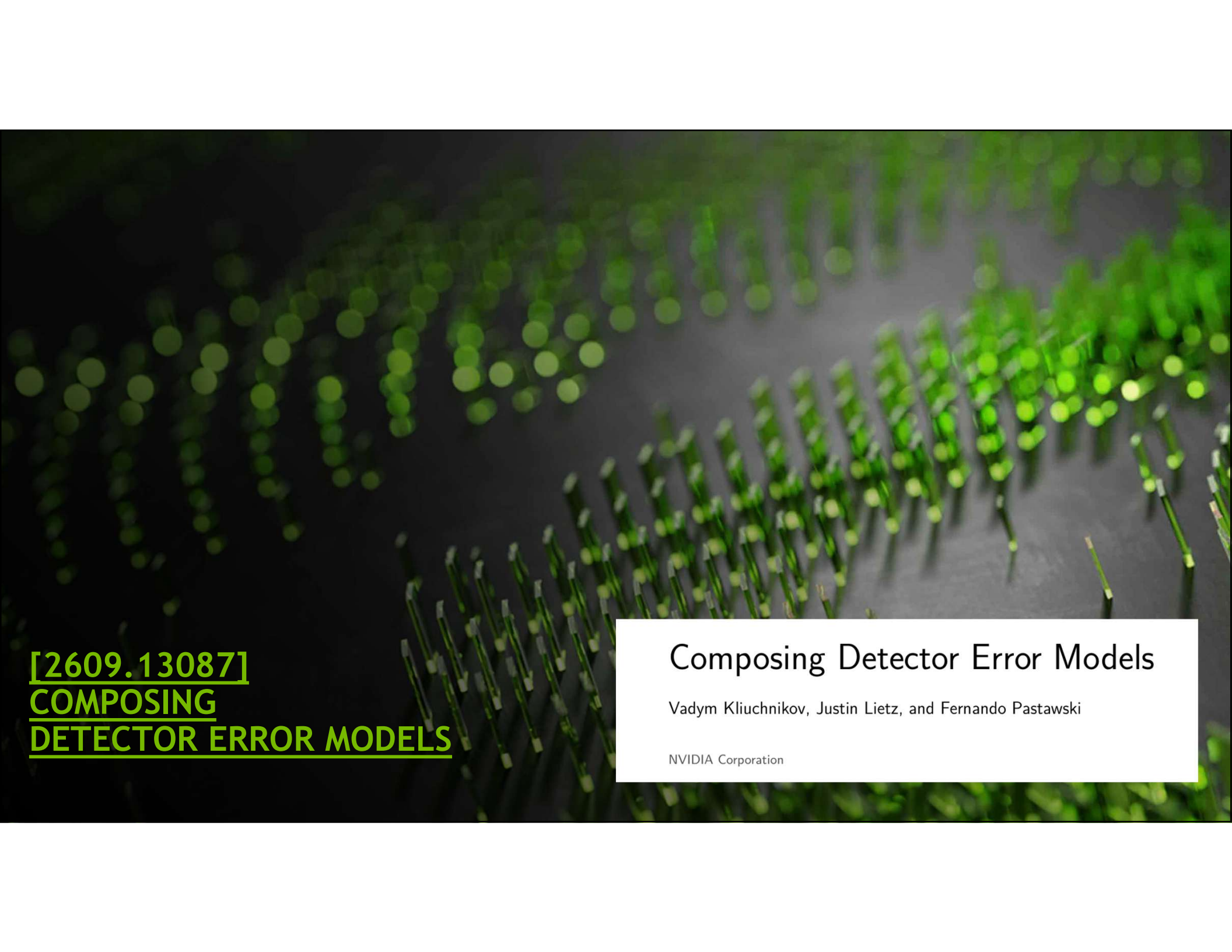




[2609.13087]
COMPOSING
DETECTOR ERROR MODELS

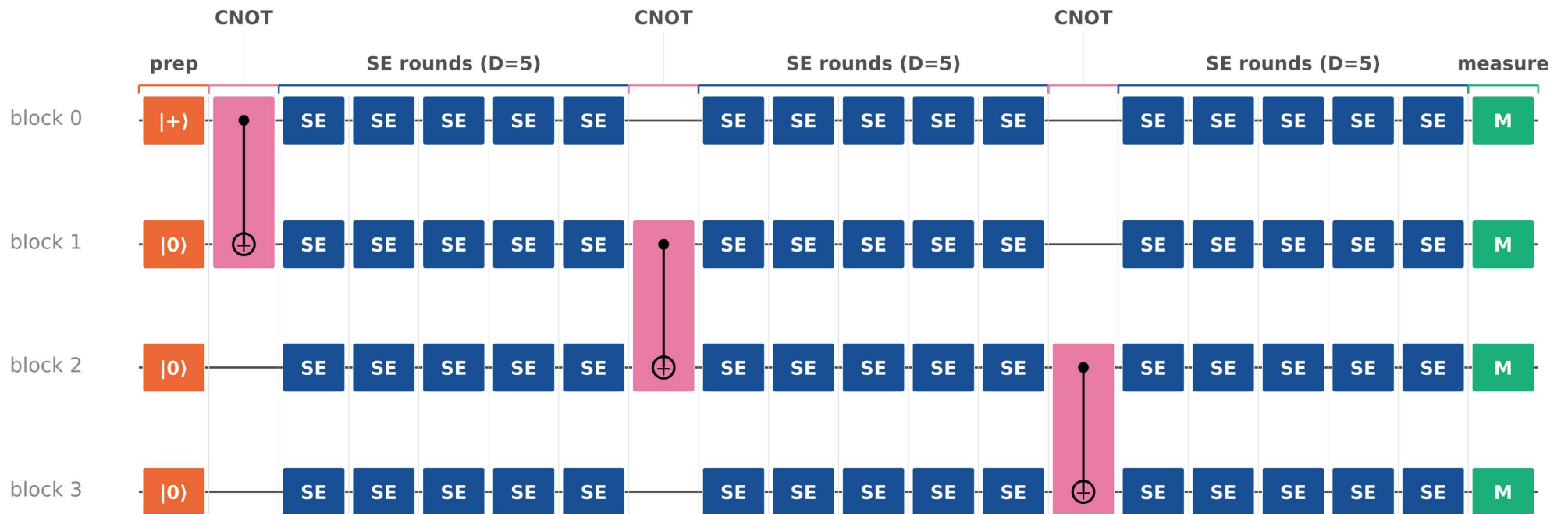
Composing Detector Error Models

Vadym Kliuchnikov, Justin Lietz, and Fernando Pastawski

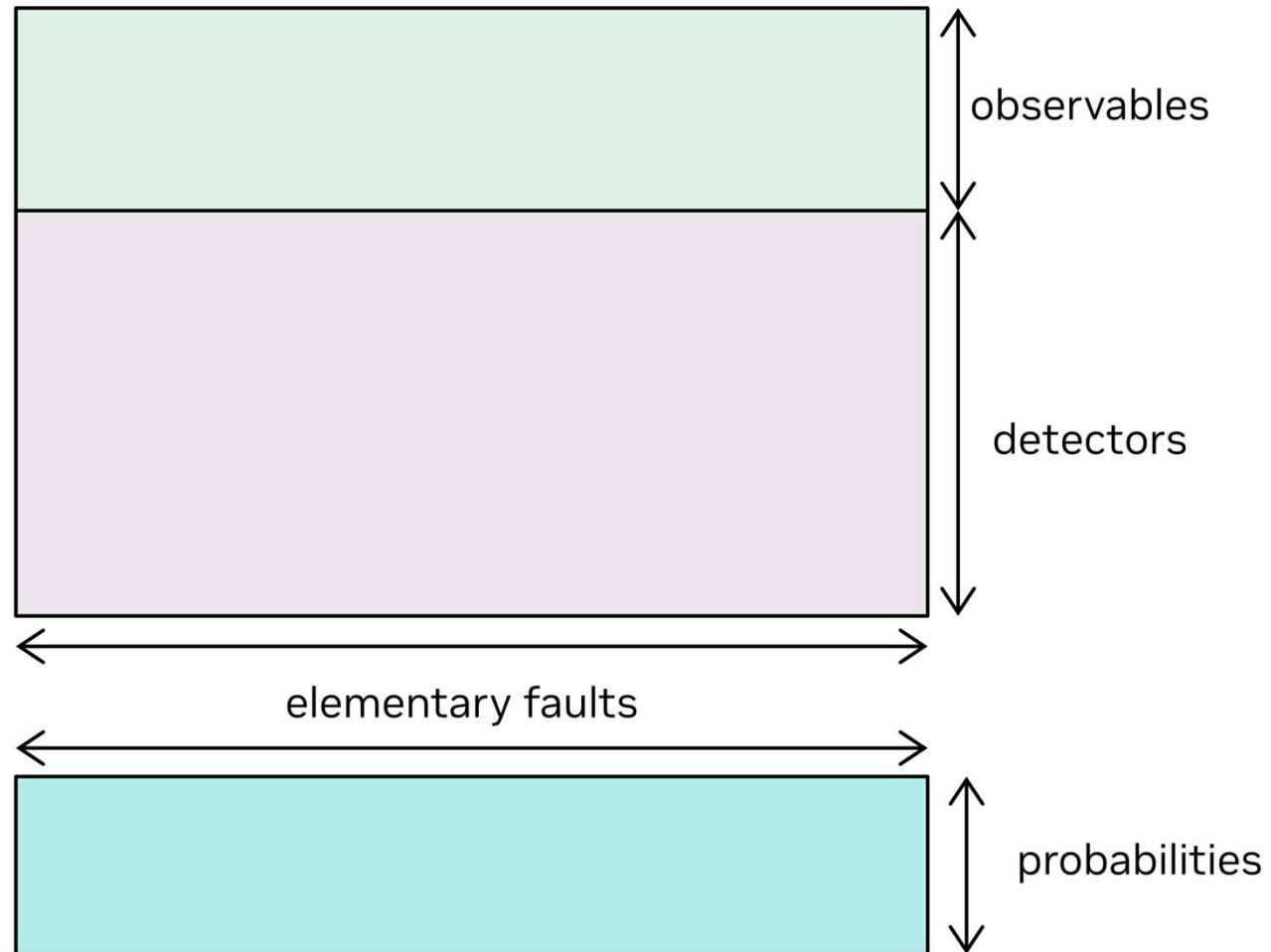
NVIDIA Corporation

Cat-state (GHZ) circuit — surface code $d=5$, 4 logical qubits, $D=5$ SE rounds

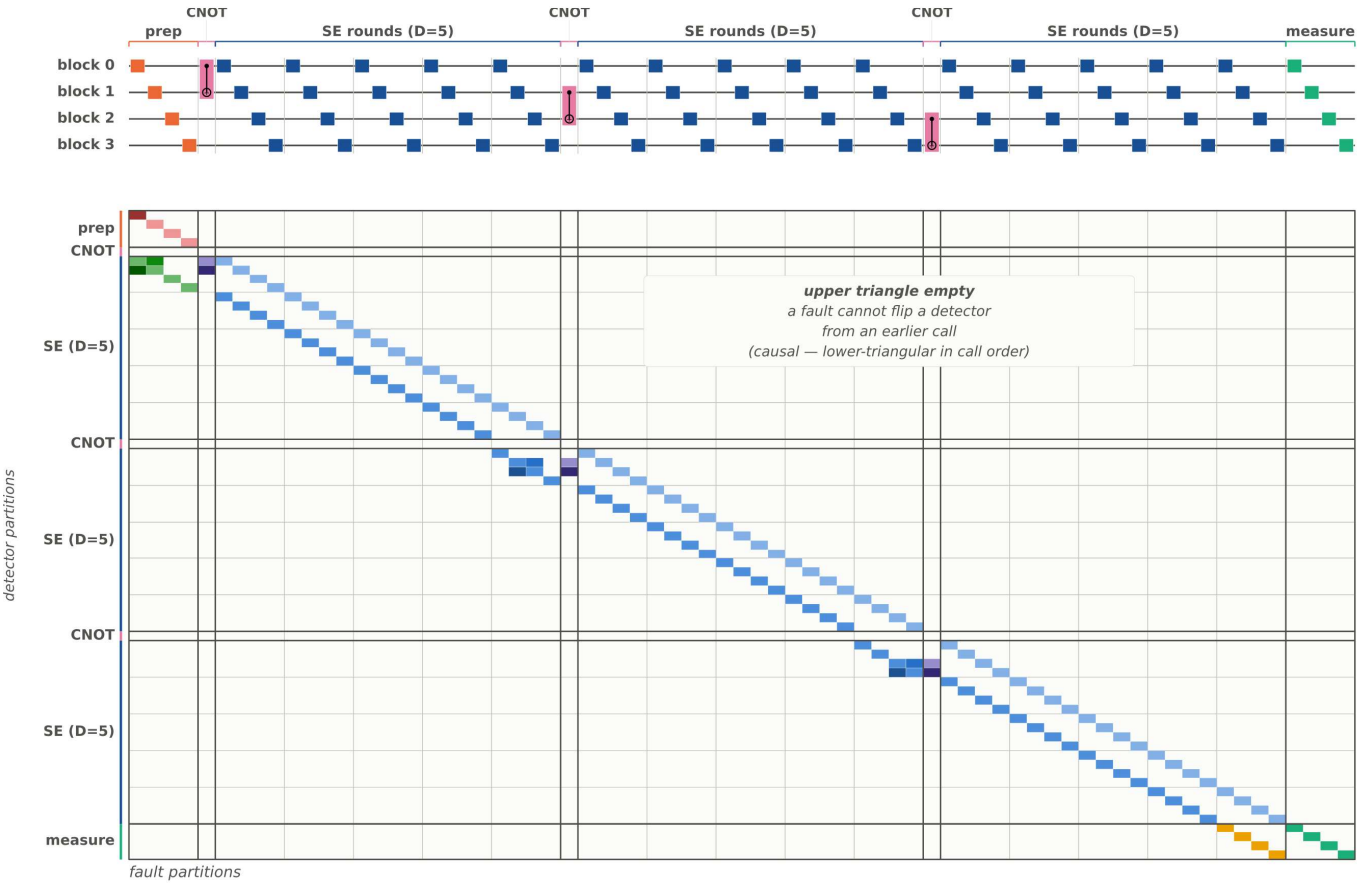
Each SE round is one time-slice: all blocks' syndrome-extraction gadgets run in parallel (aligned in a column). Block i controls block $i+1$; the head block is prepared in $|+\rangle$, the rest in $|0\rangle$, then all blocks are destructively measured.



DEM as bitmatrix



Symbolic DEM — cat state (GHZ), surface code d=5, 4 logical qubits, D=5 SE rounds



Gadgets (row / col partitions)

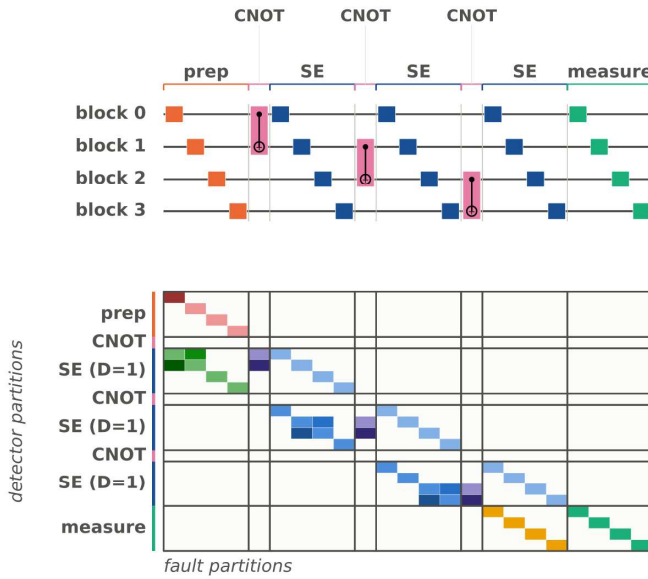
gadget	det	fault
prep	36	985
cnot	0	150
se	24	576
measure	12	25

Distinct blocks (det-row × flt-col)

block class	size	n	polynomial
prep-det × prep-flt	36×985	3	D
prep-det × prep-flt	36×985	1	H
se-det × prep-flt	24×985	4	G
se-det × prep-flt	24×985	1	V·G
se-det × prep-flt	24×985	1	X·G
se-det × cnot-flt	24×150	3	W
se-det × cnot-flt	24×150	3	Y
se-det × se-flt	24×576	60	A
se-det × se-flt	24×576	56	A + C
se-det × se-flt	24×576	2	V·A + V·C
se-det × se-flt	24×576	2	X·A + X·C
measure-det × se-flt	12×576	4	J·A + J·C
measure-det × measure-flt	12×25	4	K

1632 detectors × 39050 fault mechanisms, arranged as a 71×71 grid of blocks (144 non-zero).
Each **column** is one gadget call's fault partition; each **row** is one call's detector partition. A filled block means a fault in that column flips a detector in that row.

Symbolic DEM — cat state (GHZ), surface code d=5, 4 logical qubits, D=1 SE rounds



Gadgets (row / col partitions)

gadget	det	fault
prep	36	985
cnot	0	150
se	24	576
measure	12	25

Distinct blocks (det-row × flt-col)

block class	size	n	polynomial
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se-det × prep-flt	24×985	4	G
se-det × prep-flt	24×985	1	V·G
se-det × prep-flt	24×985	1	X·G
se-det × cnot-flt	24×150	3	W
se-det × cnot-flt	24×150	3	Y
se-det × se-flt	24×576	12	A
se-det × se-flt	24×576	8	A + C
se-det × se-flt	24×576	2	V·A + V·C
se-det × se-flt	24×576	2	X·A + X·C
measure-det × se-flt	12×576	4	J·A + J·C
measure-det × measure-flt	12×25	4	K

480 detectors × 11402 fault mechanisms, arranged as a 23×23 grid of blocks (48 non-zero).

Each **column** is one gadget call's fault partition; each **row** is one call's detector partition. A filled block means a fault in that column flips a detector in that row.

Per-gadget DEM block matrices

prep |+)

det ↓ / flt →	fault
det	H
outSyn ₀	G
outVdet ₀	0

prep |0)

det ↓ / flt →	fault
det	D
outSyn ₀	G
outVdet ₀	0

SE

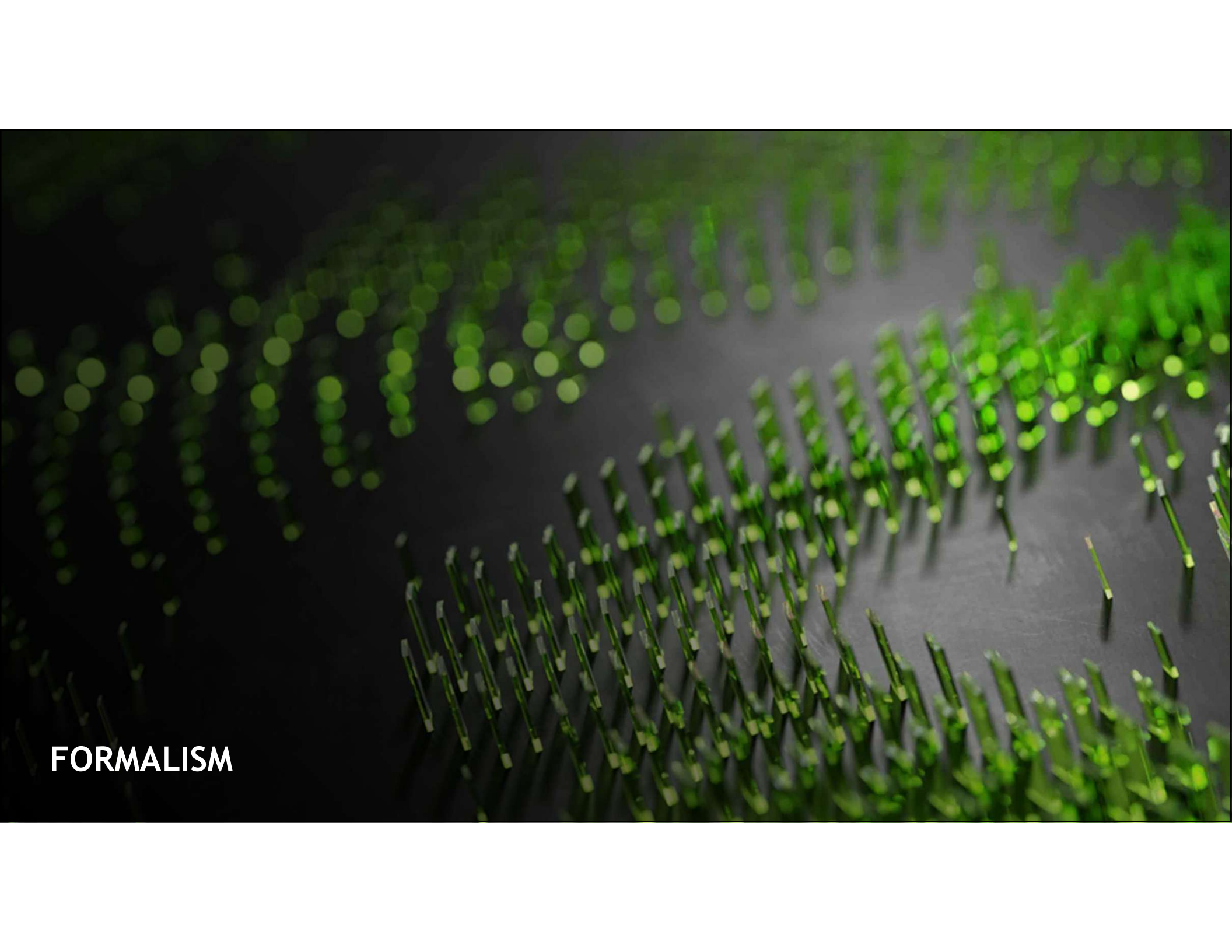
det ↓ / flt →	inSyn ₀	inVdet ₀	fault
det	1	1	A
outSyn ₀	1	0	C
outVdet ₀	1	0	A

CNOT

det ↓ / flt →	inSyn ₀	inSyn ₁	inVdet ₀	inVdet ₁	fault
outSyn ₀	1	V	0	0	W
outSyn ₁	X	1	0	0	Y
outVdet ₀	0	0	1	V	0
outVdet ₁	0	0	X	1	0

measure Z

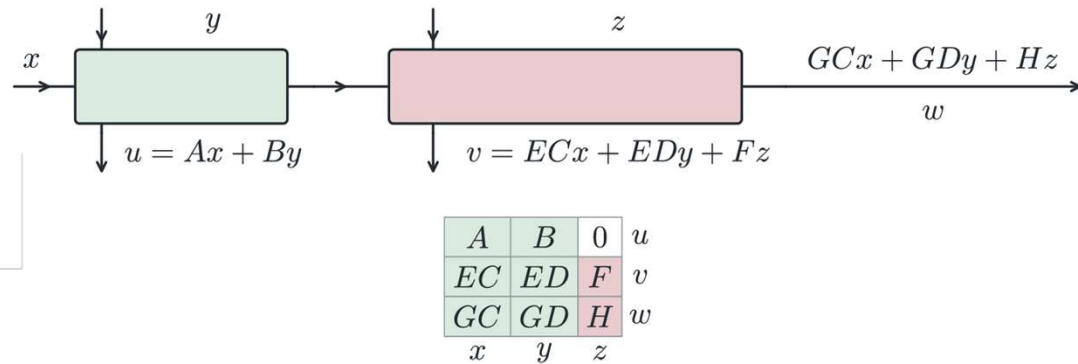
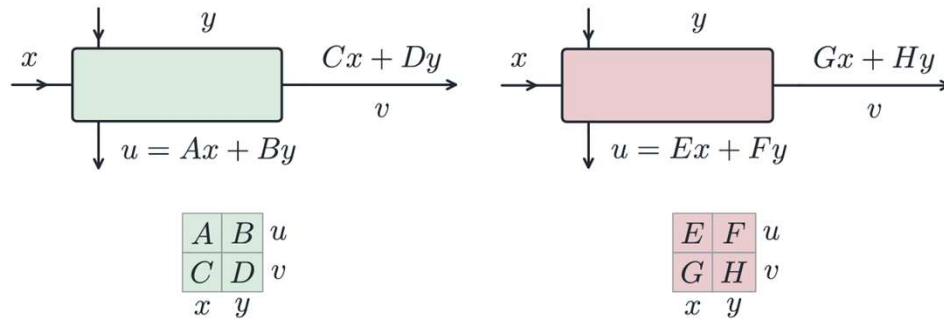
det ↓ / flt →	inSyn ₀	inVdet ₀	fault
det	J	J	K



FORMALISM

“TENSOR NETWORKS” FOR BIT MATRICES

(direct sum networks)



Towards an algebraic theory of Boolean circuits

Yves Lafont [✉](#) [Show more](#) [v](#)

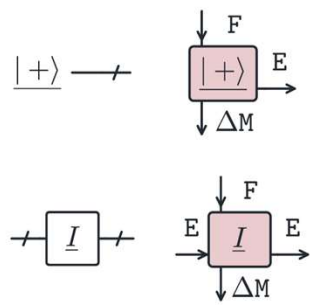
Journal of Pure and Applied Algebra

Date: 1 November 2003

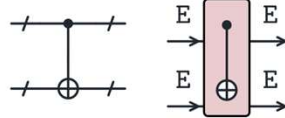
Pages: 257-310

Volume: [Volume 184, Issues 2–3](#)

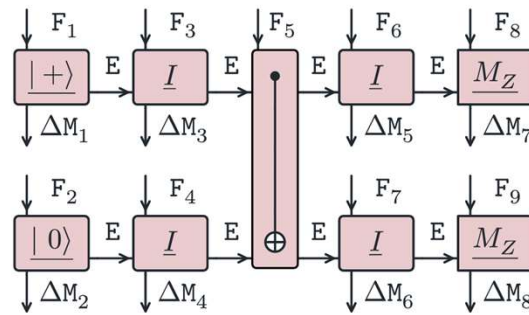
FAULT PROPAGATION AS A DIAGRAM



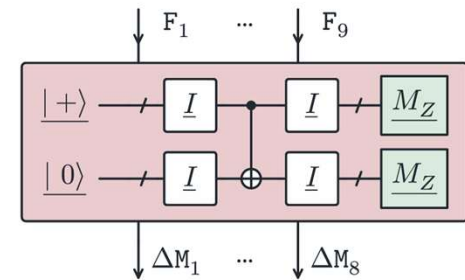
(a)



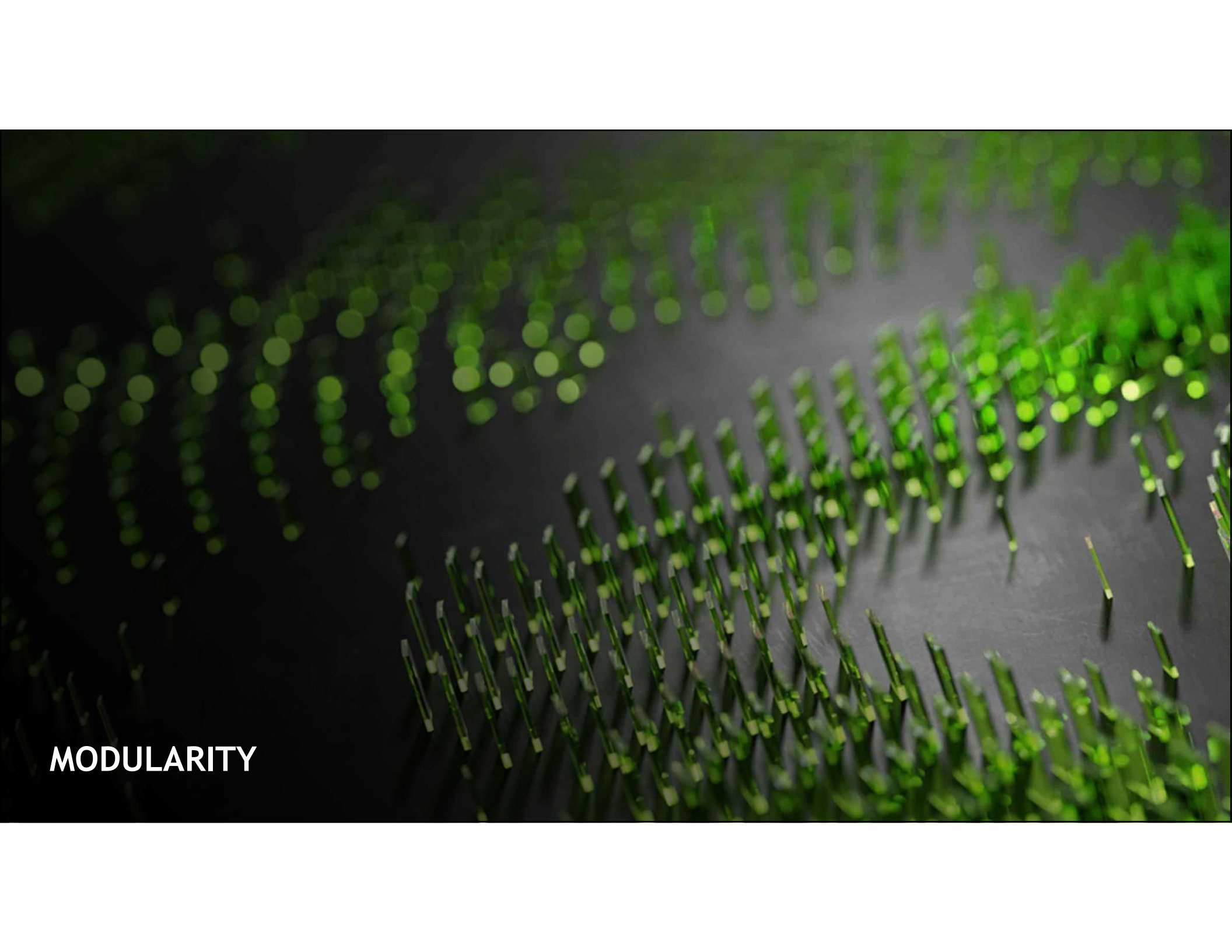
(b)



(c)

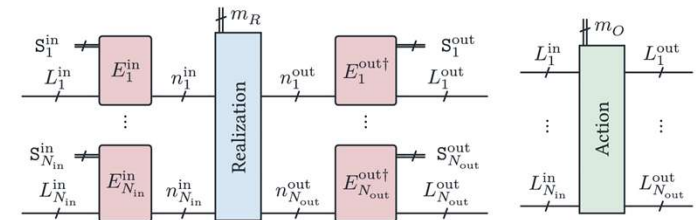
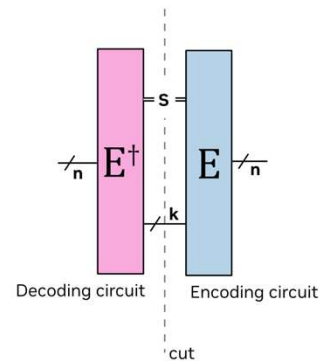
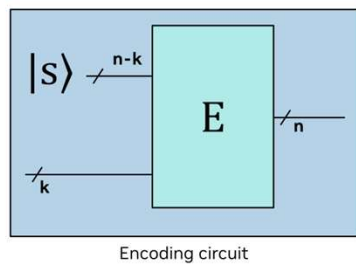
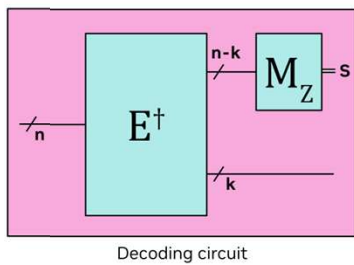
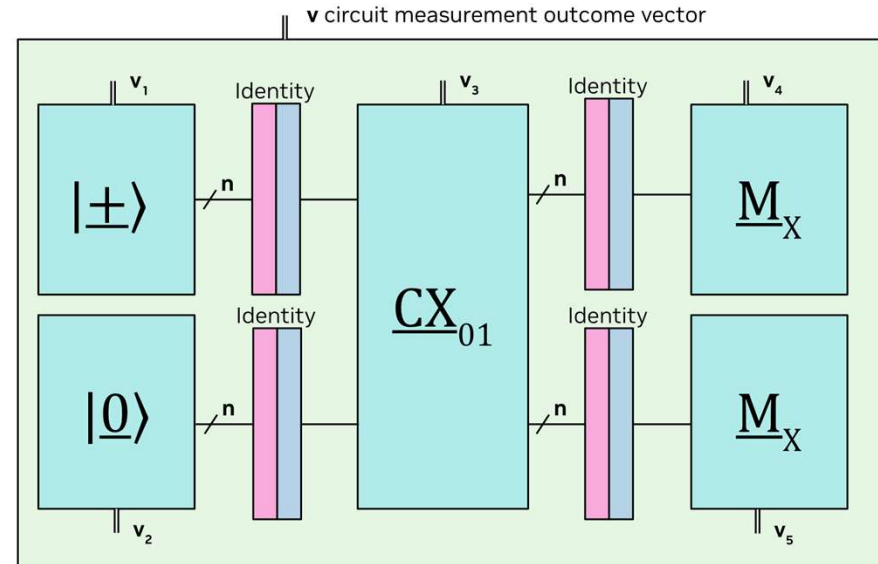
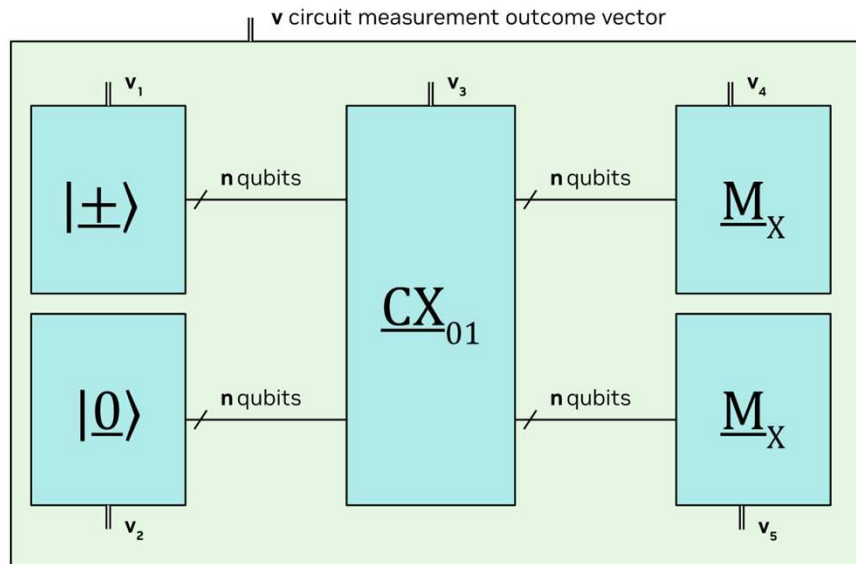


(d)



MODULARITY

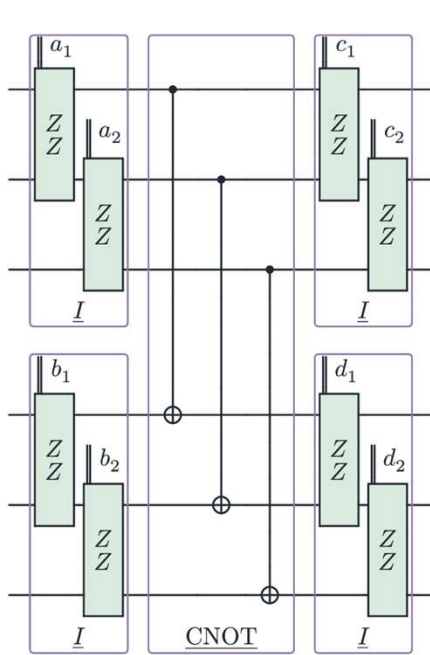
HOW DO WE CUT CIRCUIT INTO PIECES ?



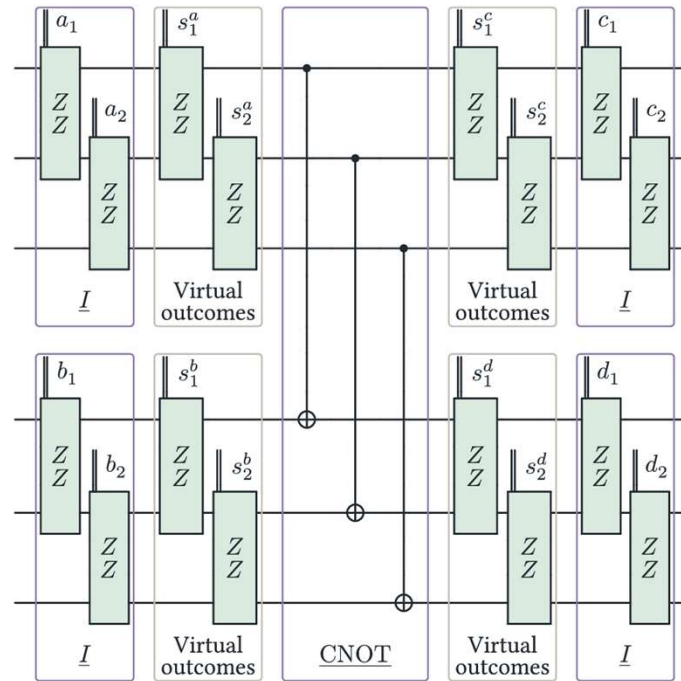
- $\mathbf{E}$ is $[n,k]$ QEC code encoding Clifford:
- maps first k X,Z pairs to logical operators
 - the rest $n-k$ of Z operator to code stabilizer

$\mathbf{s}$ is code syndrome

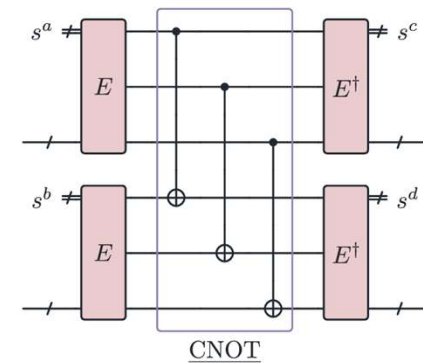
CUT BOUNDARIES VIA ENCODING CIRCUITS



(a)



(b)



(c)

Fault tolerance of stabilizer channels

Michael E. Beverland,^{1,*} Shilin Huang,^{2,†} and Vadym Kliuchnikov¹

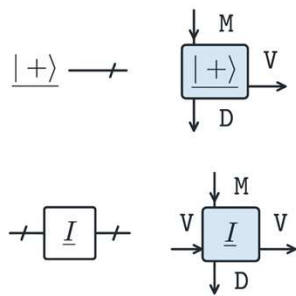
¹Microsoft Quantum, Redmond, WA 98052, USA

²Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708, USA

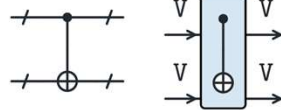
[EDEM] Section 3.1 : Formal results on cutting detectors

DEQ <https://github.com/microsoft/qdk-ec/tree/main/deq>

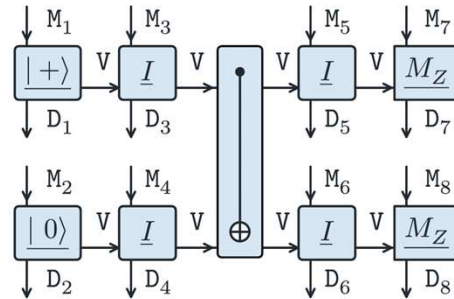
EXTENDED DETECTOR MAP



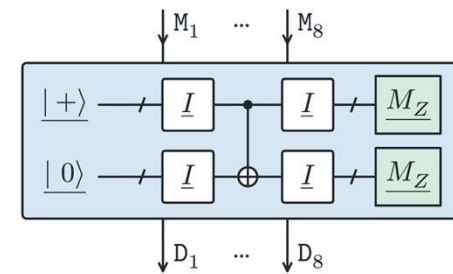
(a)



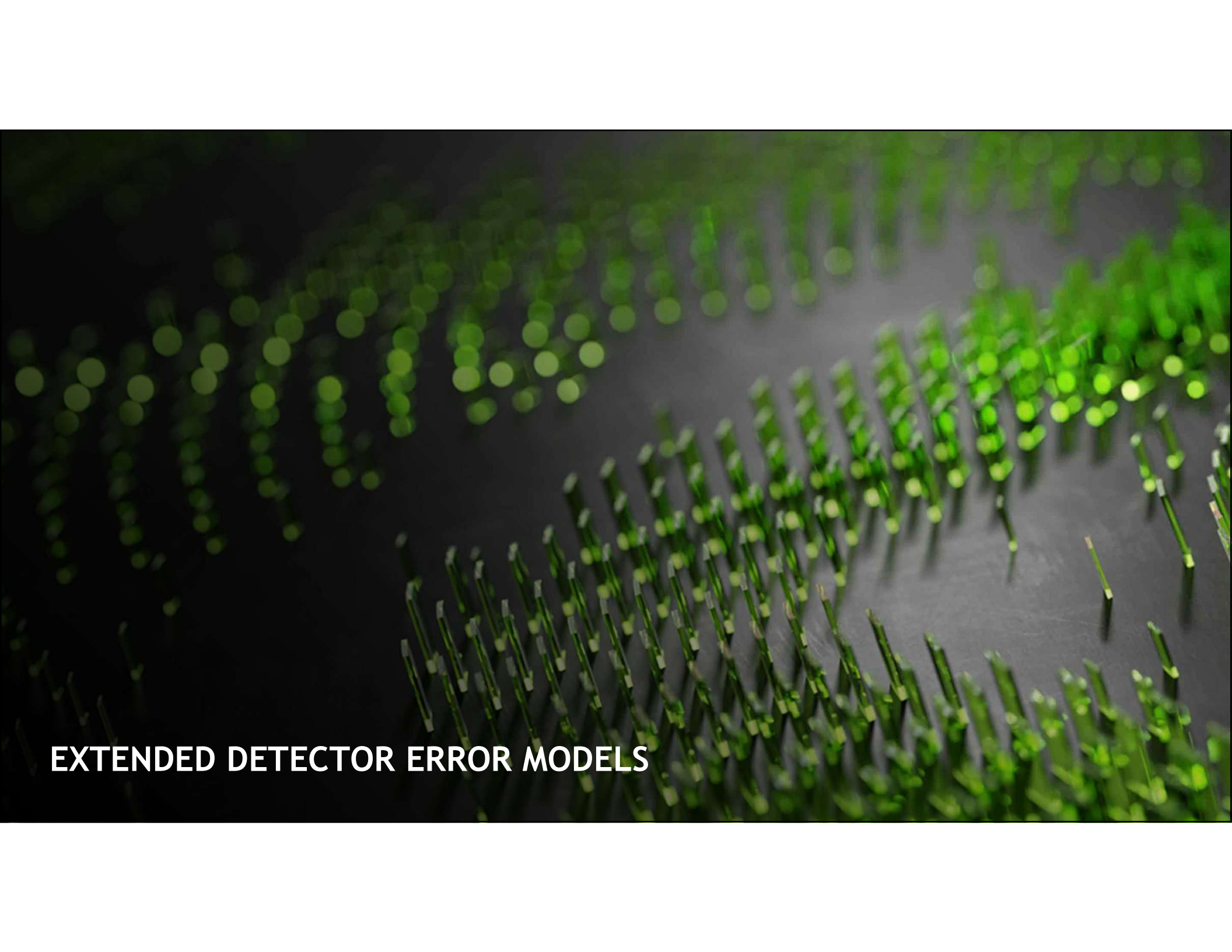
(b)



(c)

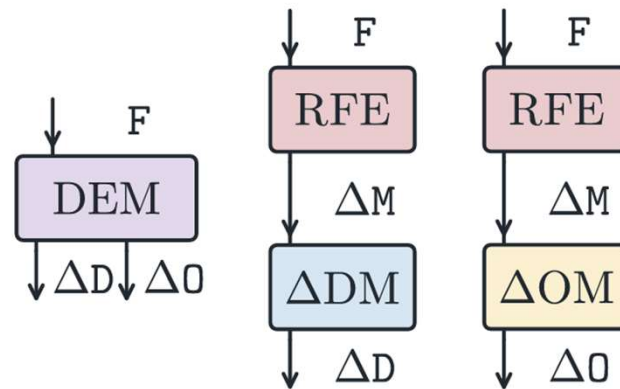


(d)

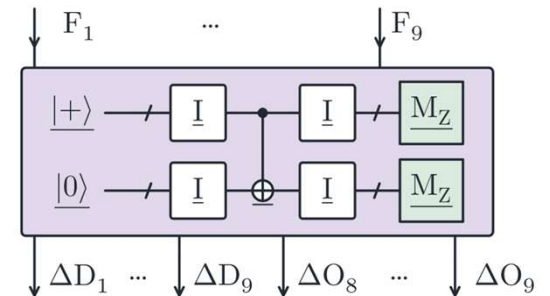
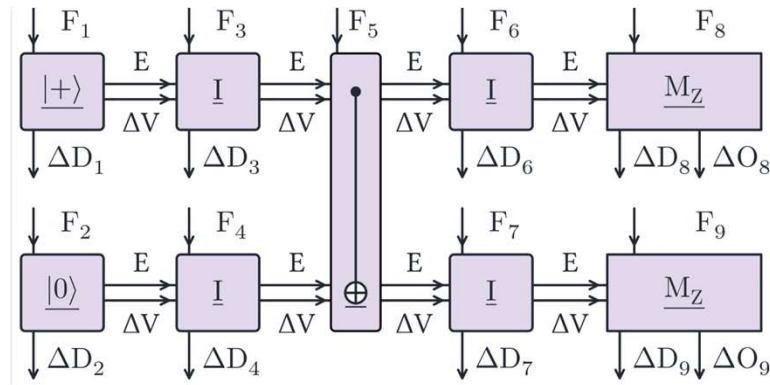
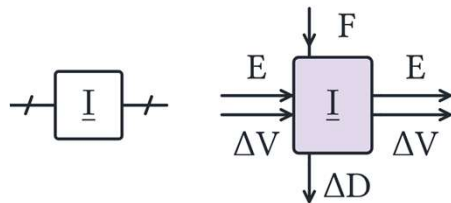
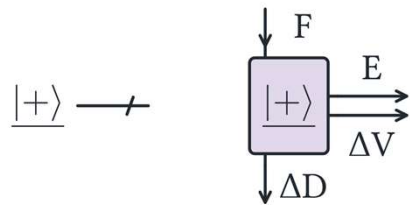
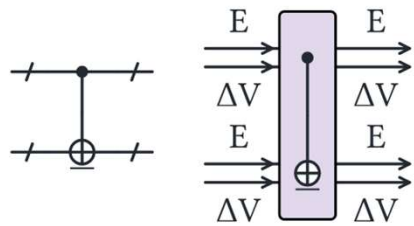


EXTENDED DETECTOR ERROR MODELS

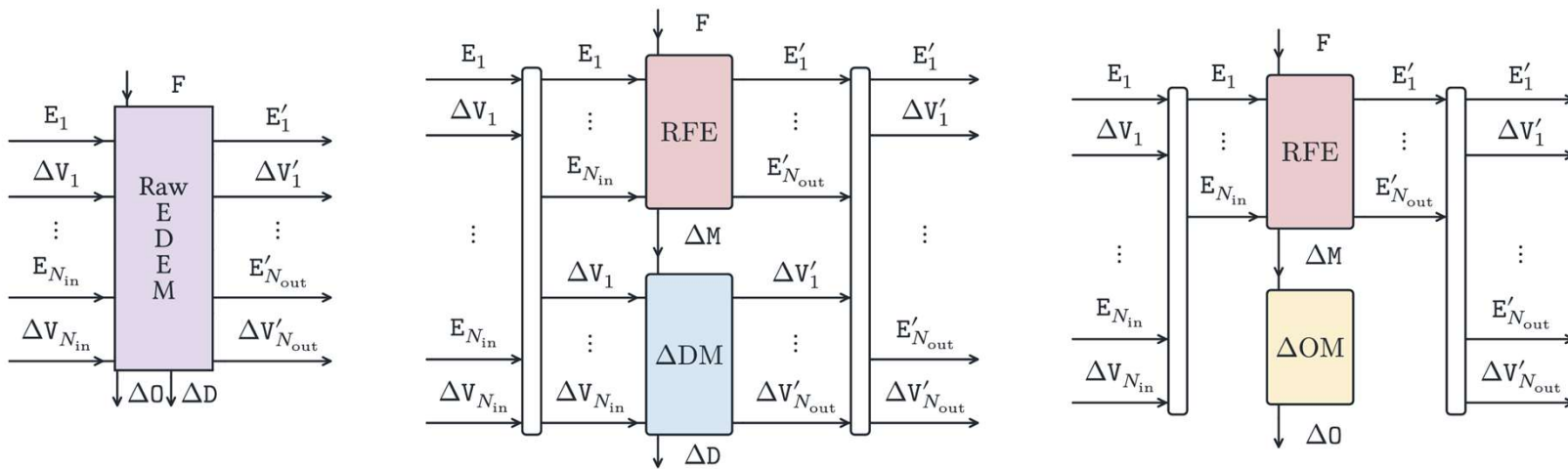
DETECTOR ERROR MODEL AS A MAP



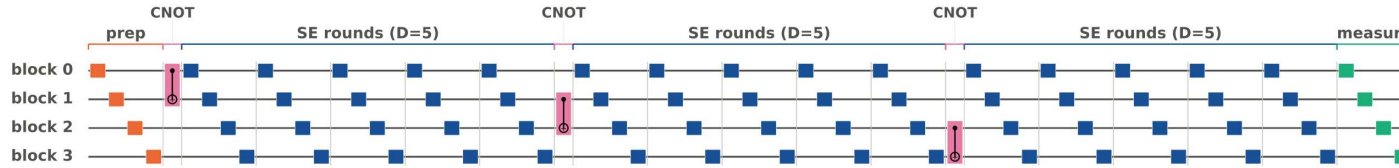
DEM FOR BELL EXPERIMENT



EXTENDED DETECTOR ERROR MODELS

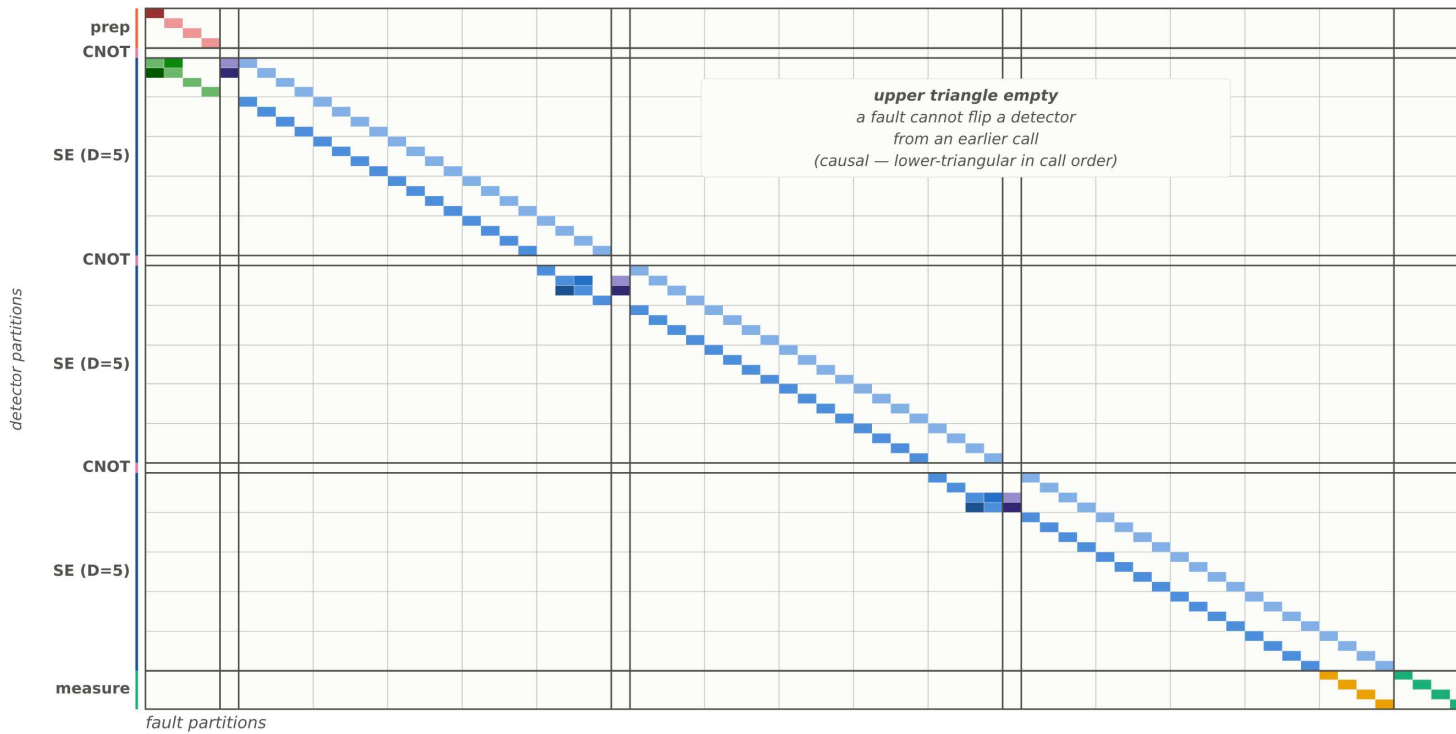


Symbolic DEM — cat state (GHZ), surface code d=5, 4 logical qubits, D=5 SE rounds



Gadgets (row / col partitions)

gadget	det	fault
prep	36	985
cnot	0	150
se	24	576
measure	12	25



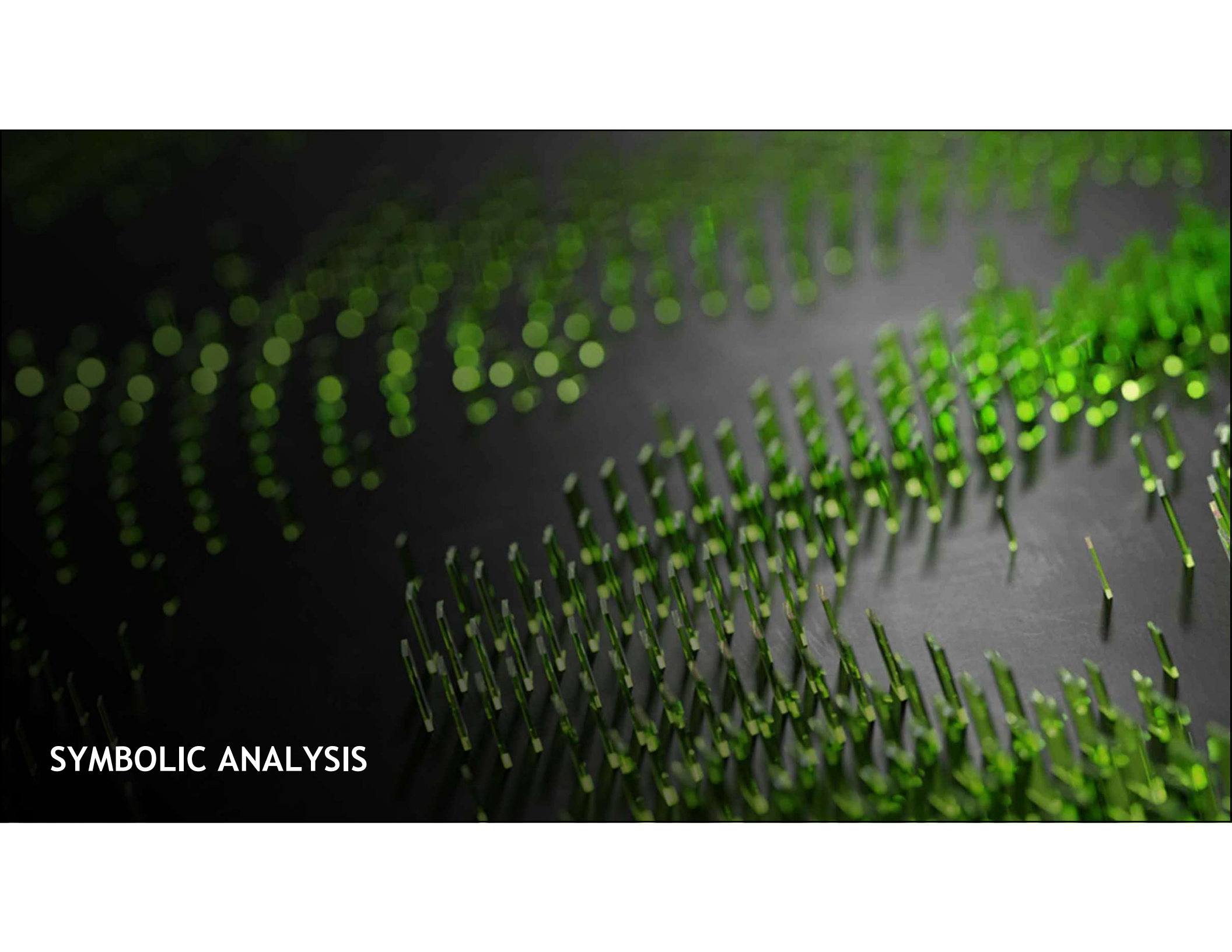
Distinct blocks (det-row × flt-col)

block class	size	n	polynomial
prep-det × prep-flt	36×985	3	D
prep-det × prep-flt	36×985	1	H
se-det × prep-flt	24×985	4	G
se-det × prep-flt	24×985	1	V·G
se-det × prep-flt	24×985	1	X·G
se-det × cnot-flt	24×150	3	W
se-det × cnot-flt	24×150	3	Y
se-det × se-flt	24×576	60	A
se-det × se-flt	24×576	56	A + C
se-det × se-flt	24×576	2	V·A + V·C
se-det × se-flt	24×576	2	X·A + X·C
measure-det × se-flt	12×576	4	J·A + J·C
measure-det × measure-flt	12×25	4	K

1632 detectors × 39050 fault

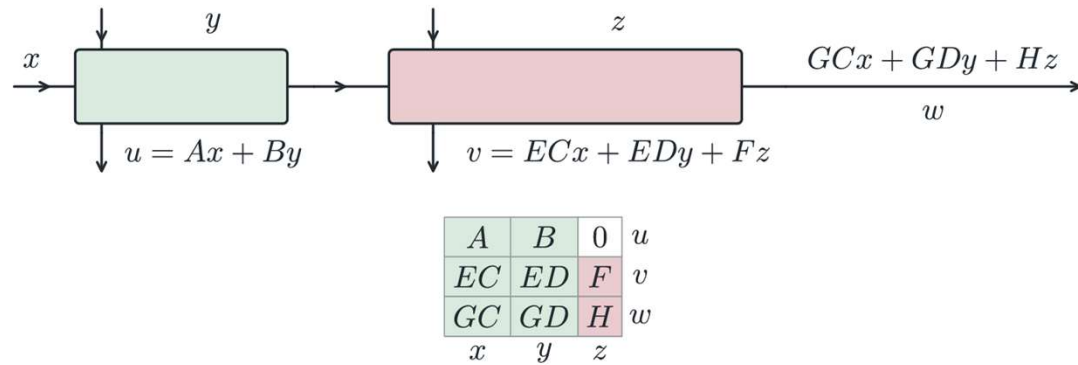
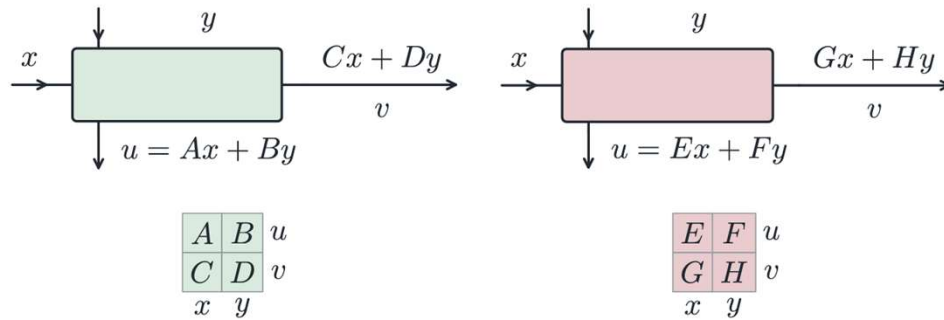
mechanisms, arranged as a 71×71 grid of blocks (144 non-zero).

Each **column** is one gadget call's fault partition; each **row** is one call's detector partition. A filled block means a fault in that column flips a detector in that row.

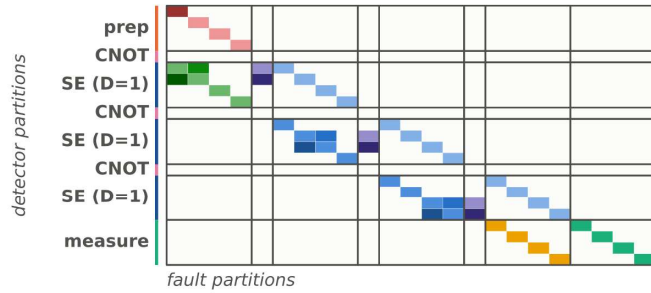
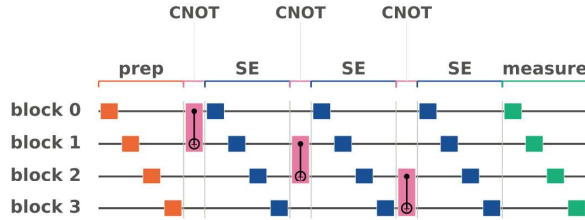


SYMBOLIC ANALYSIS

DIAGRAM COMPOSITION AND MATRIX POLYNOMIALS



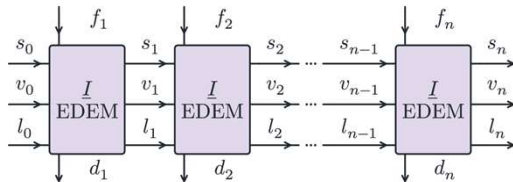
Symbolic DEM — cat state (GHZ), surface code d=5, 4 logical qubits, D=1 SE rounds



$$v_k = D_F f_k + s_0 + S_F f_{k-1}^\Sigma, f_k^\Sigma = \sum_{j=1}^k f_j \quad (31)$$

$$s_k = s_0 + S_F f_k^\Sigma, l_k = l_0 + L_F f_k^\Sigma \quad (32)$$

$$d_k = D_F f_k + (D_F + S_F) f_{k-1}, \quad (k \geq 2) \quad (33)$$



Gadgets (row / col partitions)

gadget	det	fault
prep	36	985
cnot	0	150
se	24	576
measure	12	25

Distinct blocks (det-row × flt-col)

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se-det × cnot-flt	24×150	3	W
se-det × cnot-flt	24×150	3	Y
se-det × se-flt	24×576	12	A
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480 detectors × 11402 fault mechanisms, arranged as a 23×23 grid of blocks (48 non-zero).

Each **column** is one gadget call's fault partition; each **row** is one call's detector partition. A filled block means a fault in that column flips a detector in that row.

Per-gadget DEM block matrices

prep |+)

det ↓ / flt →	fault
det	H
outSyn ₀	G
outVdet ₀	0

prep |0)

det ↓ / flt →	fault
det	D
outSyn ₀	G
outVdet ₀	0

SE

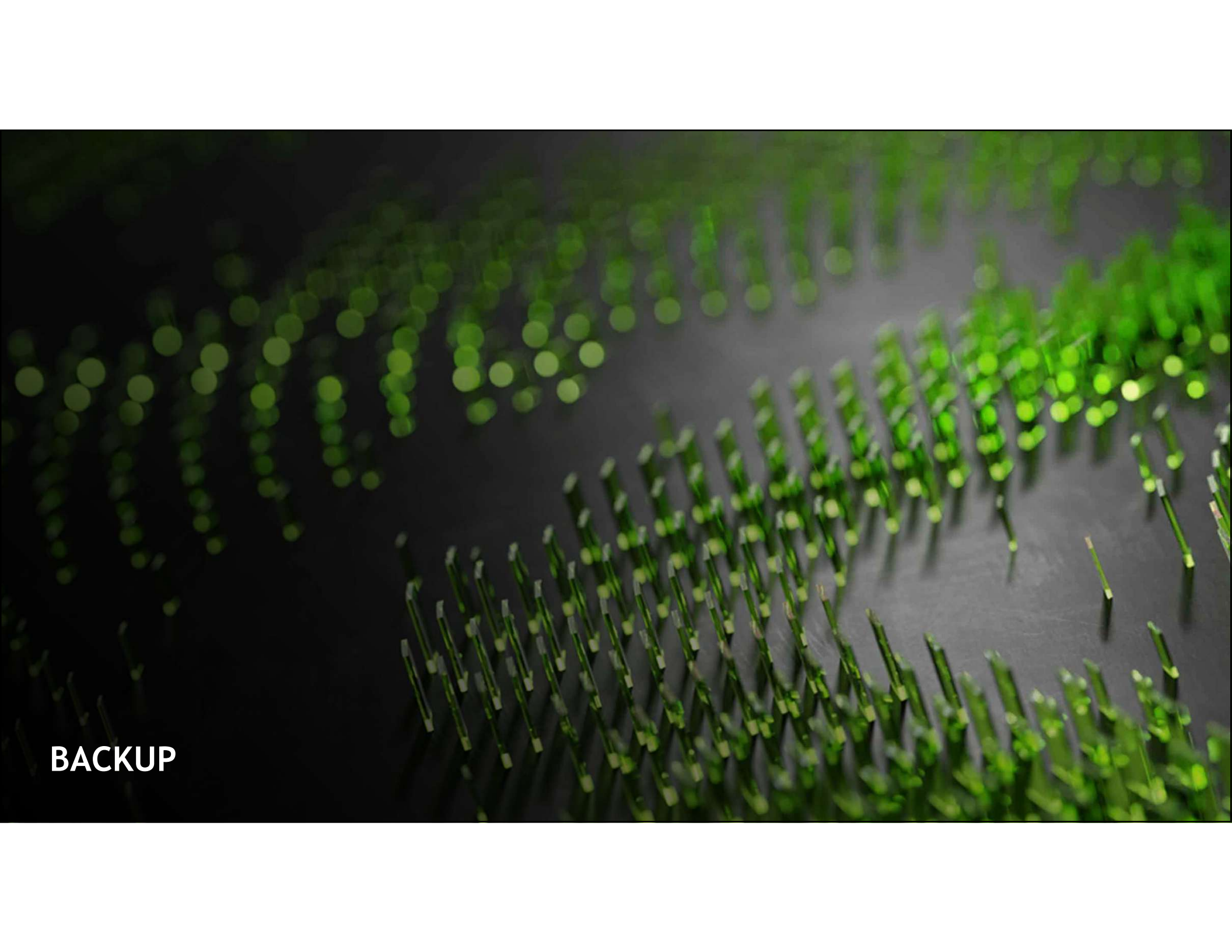
det ↓ / flt →	inSyn ₀	inVdet ₀	fault
det	1	1	A
outSyn ₀	1	0	C
outVdet ₀	1	0	A

CNOT

det ↓ / flt →	inSyn ₀	inSyn ₁	inVdet ₀	inVdet ₁	fault
outSyn ₀	1	V	0	0	W
outSyn ₁	X	1	0	0	Y
outVdet ₀	0	0	1	V	0
outVdet ₁	0	0	X	1	0

measure Z

det ↓ / flt →	inSyn ₀	inVdet ₀	fault
det	J	J	K



BACKUP

Fault compressor

Many elementary faults act identically on the DEM — same detectors, same observables, same residual error. They are indistinguishable to the decoder. The **fault compressor** maps the full elementary-fault set F onto a representative subset collapsing each class of equivalent faults to a single representative.



Fault compressor. A linear map onto a representative subset of the elementary faults.

Compression identity

Wiring the compressor in front of the **compressed DEM** — the same DEM rebuilt on the smaller fault set F — reproduces the original gadget DEM exactly. Compression re-parametrizes the fault input; it leaves the $(V, S, L) \rightarrow (O, B, D, \dots)$ map untouched.

$$\text{DEM} = \text{Compressor} \rightarrow \text{DEM}'$$

